

Navier--Stokes Existence or Breakdown

Notes from a lecture by Javier Gómez-Serrano

These notes started as an AI-cleaned transcript of the talk.

1. Introduction

This lecture concerns the Navier–Stokes existence or breakdown problem, one of the seven Millennium Prize Problems posed by the Clay Mathematics Institute around the year 2000.

The problem asks, roughly:

given smooth initial data for the 3D incompressible Navier–Stokes equations,

- do smooth solutions exist for all time?
- or can singularities form in finite time?

The talk emphasizes that, although the Clay problem remains open, the field has seen substantial activity and major conceptual progress in recent years.

2. Historical background

2.1. Leonardo da Vinci and early observations

Leonardo da Vinci studied fluid motion and apparently coined the term *turbulence* (Italian: *turbolenza*). His observations already captured a multiscale picture of fluid flow: large eddies feeding into smaller ones.

He wrote:

Observe the motion of the surface of the water, which resembles that of hair. The water has eddying motions, one part of which is due to the principal current and the other to the random and reverse motion.

So, already around 1500, there was a qualitative awareness of the cascade-like structure of fluid motion.

2.2. Bernoulli, d'Alembert, Euler, Prandtl

A few milestones:

- **Bernoulli** (Daniel Bernoulli, 1700–1782) related higher fluid velocity to lower pressure, leading to explanations such as lift on airplane wings. He published **Hydrodynamica** in 1738.
- **d'Alembert** (Jean le Rond d'Alembert, 1717–1783) considered drag and argued that an inviscid fluid should produce zero drag, in contradiction with physical observations (1752). This is the famous *d'Alembert paradox*.
- **Euler** (Leonhard Euler, 1707–1783) wrote the incompressible Euler equations (1757), describing ideal fluids without viscosity.
- **Prandtl** (Ludwig Prandtl, 1875–1953) explained the paradox through *boundary layer theory* (1904): even very small viscosity can produce substantial drag near boundaries.

Thus the mathematical theory of fluid mechanics emerged from a combination of physical observation and formal modeling.

3. The Euler and Navier–Stokes equations

Let $u = u(x, t)$ denote the velocity field, $p = p(x, t)$ the pressure, ρ the density, ν the viscosity, and $f = f(x, t)$ an external force.

The incompressible fluid equations are

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f$$

together with the incompressibility condition

$$\nabla \cdot u = 0.$$

- If $\nu = 0$, one obtains the incompressible Euler equations.
- If $\nu > 0$, one obtains the incompressible Navier–Stokes equations.

These equations express Newton's law $F = ma$ for a fluid parcel.

3.1. The equations

The incompressible Navier–Stokes equations above combine the local acceleration $\partial_t u$, the nonlinear advection $(u \cdot \nabla)u$, the pressure gradient ∇p , and the viscous diffusion $\nu \Delta u$. Together they govern the evolution of a fluid velocity field.

3.2. The nonlinear term

The main mathematical difficulty comes from the nonlinear advection term

$$(u \cdot \nabla)u.$$

This term means that the velocity field affects its own evolution. The equation is therefore nonlinear, and this raises the possibility of dramatic phenomena such as growth of gradients or formation of singularities.

3.3. Incompressibility

The condition

$$\nabla \cdot u = 0$$

means the fluid is incompressible: parcels preserve volume. This prevents simple compression-type blowup mechanisms that might occur in compressible flow.

4. The Clay Millennium problem

The 3D incompressible Navier–Stokes problem, in one standard formulation, is:

Given smooth divergence-free initial data u_0 on either $\mathbb{R}^3$ or the torus $\mathbb{T}^3$, consider the evolution

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u \quad \nabla \cdot u = 0 \quad u(\cdot, 0) = u_0.$$

The question is whether one of the following holds:

1. **Global regularity:** for every smooth initial datum, there exists a smooth solution for all time.
2. **Finite-time breakdown:** there exists smooth initial datum for which a singularity forms in finite time.

A singularity means that some norm of the solution, typically involving u or its derivatives, becomes unbounded in finite time.

5. Three broad scenarios

The speaker described three conceptual possibilities:

1. **Global smoothness and uniqueness**

Smooth solutions remain smooth for all time.

2. Singularity formation, but uniqueness persists

Singularities may form, yet the evolution remains uniquely determined in an appropriate sense.

3. Singularity formation and nonuniqueness

Singularities occur, and beyond them there may be multiple solutions.

The Clay problem is usually presented as distinguishing scenario 1 from scenarios 2 or 3 combined.

A different emphasis, associated with Ladyzhenskaya, is to distinguish uniqueness from nonuniqueness.

6. Weak solutions

To discuss global existence, one introduces weaker notions of solution.

6.1. Definition of weak solution

A **weak solution** is a function u that satisfies the Navier–Stokes equations only after testing against smooth divergence-free test functions.

For simplicity, consider the unforced equation on $\mathbb{R}^3$ or $\mathbb{T}^3$:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u \quad \nabla \cdot u = 0.$$

Then u is a weak solution if for every smooth compactly supported divergence-free test field φ ,

$$\int_0^\infty \int (u \cdot \partial_t \varphi + (u \otimes u) : \nabla \varphi + \nu \nabla u : \nabla \varphi) dx dt + \int u_0(x) \cdot \varphi(x, 0) dx = 0.$$

This is obtained by multiplying the PDE by φ and integrating by parts.

A strong solution is always a weak solution, but the converse need not hold.

6.2. Leray–Hopf weak solutions

A **Leray–Hopf weak solution** is a weak solution that also satisfies the energy inequality

$$\frac{1}{2} \|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds \leq \frac{1}{2} \|u_0\|_{L^2}^2$$

for all t .

This means the kinetic energy is nonincreasing, except for dissipation through viscosity.

6.3. Leray–Hopf theorem

Leray (1934) proved the foundational existence result for 3D incompressible Navier–Stokes: for any divergence-free initial data $u_0 \in L^2(\mathbb{R}^3)$, there exists at least one global weak solution

$$u \in L_t^\infty L_x^2 L_t^2 \dot{H}_x^1$$

satisfying the energy inequality for all t . Moreover, the solution is smooth except possibly on a small exceptional set; in particular, the set of singular times has zero $\frac{1}{2}$ -dimensional Hausdorff measure.¹

Hopf (1951) later extended this to bounded domains with no-slip boundary conditions.²

Thus:

- global weak solutions exist for any L^2 initial data;

¹J. Leray, “Sur le mouvement d’un liquide visqueux emplissant l’espace,” **Acta Math.** 63, 193–248 (1934). DOI Zbl

²E. Hopf, “Über die Anfangswertaufgabe für die hydrodynamischen Grundgleichungen,” **Math. Nachr.** 4, 213–231 (1951). DOI Zbl

- but it is unknown whether they remain smooth or unique in full generality.

If one could prove that every Leray–Hopf weak solution is actually smooth, the Clay problem would be solved in the positive direction.

7. The two-dimensional case

In two spatial dimensions, the situation is much better.

7.1. Ladyzhenskaya’s theorem

Ladyzhenskaya (1958, 1969) proved global well-posedness for 2D incompressible Navier–Stokes: for any divergence-free $u_0 \in L^2(\mathbb{R}^2)$, there exists a unique global solution

$$u \in C([0, \infty); L^2) L^2_{\text{loc}}((0, \infty); H^1)$$

that is smooth for all $t > 0$ and depends continuously on the initial data.³

The key point is that certain inequalities, now often called Ladyzhenskaya inequalities, provide a stronger control in two dimensions than in three. In 2D, the inequality

$$\|u\|_{L^4} \leq C \|u\|_{L^2}^{\frac{1}{2}} \|\nabla u\|_{L^2}^{\frac{1}{2}}$$

holds, which together with the energy inequality gives enough control to close the estimates. In 3D, the analogous inequality

$$\|u\|_{L^4} \leq C \|u\|_{L^2}^{\frac{1}{4}} \|\nabla u\|_{L^2}^{\frac{3}{4}}$$

has a higher exponent on the ∇u term, and the same strategy breaks down.

So:

- in 2D Navier–Stokes, the global regularity theory is complete;
- in 3D, it remains open.

8. Partial regularity

Even if singularities exist, can we say how large the singular set is?

8.1. Suitable weak solutions

A **suitable weak solution** is a weak solution that additionally satisfies the **local energy inequality**

$$\partial_t \left(\frac{|u|^2}{2} \right) + \operatorname{div} \left(\left(\frac{|u|^2}{2} + p \right) u \right) - \nu \Delta \left(\frac{|u|^2}{2} \right) + \nu |\nabla u|^2 \leq 0$$

in the sense of distributions. This local form of energy dissipation is a stronger condition than the global energy inequality and is essential for partial regularity arguments.

8.2. Scheffer

Scheffer (1977) proved that the singular set of a suitable weak solution has parabolic Hausdorff dimension at most $\frac{5}{3}$ (later improved to at most 2).⁴

³O. A. Ladyzhenskaya, “Solution ‘in the large’ of the nonstationary boundary value problem for the Navier–Stokes system in two space variables,” **Comm. Pure Appl. Math.** 12, 427–433 (1959). DOI Zbl also **The Mathematical Theory of Viscous Incompressible Flow**, 2nd ed., Gordon & Breach (1969).

⁴V. Scheffer, “Hausdorff measure and the Navier–Stokes equations,” **Comm. Math. Phys.** 55, 97–112 (1977). DOI Zbl

8.3. Caffarelli–Kohn–Nirenberg

Caffarelli, Kohn, and Nirenberg (1982) improved this dramatically: the one-dimensional parabolic Hausdorff measure of the singular set is zero.⁵

Concretely, this implies that the singular set S satisfies $P^1(S) = 0$, meaning its one-dimensional parabolic Hausdorff measure vanishes. In particular, S cannot contain any curve of positive length in spacetime, so possible singularities are extremely sparse.

This partial regularity result historically supported optimism for global regularity: if singularities exist, they must be highly constrained.

9. Vorticity formulation

A central derived quantity is the **vorticity**

$$\omega = \nabla \times u.$$

- In 2D, ω is effectively a scalar.
- In 3D, ω is a vector field.

Writing the equations in terms of vorticity removes the pressure from the formulation.

9.1. The 2D vorticity equation

In 2D Navier–Stokes,

$$\partial_t \omega + u \cdot \nabla \omega = \nu \Delta \omega.$$

Thus vorticity is simply transported and diffused.

This is a major reason why 2D is tractable.

9.2. The 3D vorticity equation

In 3D Navier–Stokes,

$$\partial_t \omega + u \cdot \nabla \omega = \omega \cdot \nabla u + \nu \Delta \omega.$$

The extra term

$$\omega \cdot \nabla u$$

is the **vortex stretching term**. This term can amplify vorticity and is one of the main mechanisms suspected in any possible singularity formation.

So:

- in 2D, vorticity is transported;
- in 3D, vorticity can stretch and grow.

10. Why energy estimates are not enough

The kinetic energy is the most basic conserved or dissipated quantity. But for 3D Navier–Stokes, energy is **supercritical** with respect to the natural scaling of the equations.

That means energy estimates alone do not control the norms relevant to singularity formation.

Hence any proof of global regularity must use something beyond soft energy arguments. One needs a deeper structural understanding of the nonlinearity.

⁵L. Caffarelli, R. Kohn, and L. Nirenberg, “Partial regularity of suitable weak solutions of the Navier-Stokes equations,” **Comm. Pure Appl. Math.** 35, 771–831 (1982). DOI Zbl

This is reinforced by model equations and modified systems that preserve many of the same soft features, yet can exhibit blowup.

11. What was known classically

Before modern developments, one broadly knew:

- Leray–Hopf weak solutions exist globally.
- Smooth solutions exist locally in time.
- If singularities occur, they are very sparse in spacetime.
- 2D Navier–Stokes is globally well-posed.
- 3D global smoothness remained open.

The prevailing community intuition for a long time leaned toward global regularity, though without proof.

12. Regularity criteria

A common strategy is: if one cannot prove global regularity directly, perhaps one can identify conditions under which a weak solution must be smooth.

12.1. Prodi–Serrin–Ladyzhenskaya criteria

If a Leray–Hopf weak solution lies in a mixed space

$$u \in L_t^p L_x^q$$

with

$$\frac{2}{p} + \frac{3}{q} \leq 1 \quad q > 3$$

then the solution is smooth.

The endpoint case ($q=3$) was much harder and was handled later.

12.2. Escauriaza–Seregin–Šverák

Escauriaza, Seregin, and Šverák (2003) proved the endpoint $q = 3$ case: if

$$u \in L_t^\infty L_x^3$$

then the solution is smooth.⁶ The proof uses a backward uniqueness argument for the vorticity equation: assuming blowup, one shows that the vorticity must vanish identically, leading to a contradiction.

These regularity criteria are useful because they imply that any blowup must violate them in a specific quantitative way.

12.3. Beale–Kato–Majda criterion

For Euler, and in related forms for Navier–Stokes, a basic blowup criterion says that if a smooth solution blows up at time T , then

$$\int_0^T \|\omega(\cdot, t)\|_{L^\infty} dt = \infty.$$

⁶L. Escauriaza, G. Seregin, and V. Šverák, “ $L_{3,\infty}$ -solutions of Navier–Stokes equations and backward uniqueness,” *Russ. Math. Surveys* 58, 211–250 (2003). DOI Zbl

So finite-time blowup requires vorticity to become sufficiently large.

12.4. Constantin–Fefferman criterion

Constantin and Fefferman (1993) refined this by showing that not only magnitude but also the geometry of vorticity matters.⁸ Define the vorticity direction $\xi = \frac{\omega}{|\omega|}$ wherever $\omega \neq 0$. If

$$\int_0^T \|\nabla \xi(\cdot, t)\|_{L^\infty}^2 dt < \infty$$

then the solution remains smooth up to time T . In other words, a singularity can form only if the direction field of the vorticity develops sufficiently rapid spatial oscillations. This geometric condition complements the BKM magnitude criterion and suggests that blowup, if it occurs, must be highly structured.

13. Sharp well-posedness spaces

Researchers then sought the largest function spaces in which one can still prove well-posedness.

13.1. Koch–Tataru

Koch and Tataru (2001) proved global well-posedness for small initial data in the critical space BMO^{-1} : there exists $\varepsilon > 0$ such that for any divergence-free u_0 with $\|u_0\|_{\text{BMO}^{-1}} < \varepsilon$, there exists a unique global solution

$$u \in L_t^\infty \text{BMO}^{-1} L_t^2 C^{0, \frac{1}{2}}$$

that is smooth for $t > 0$.⁹

This space is essentially optimal: it is invariant under the natural scaling of the equations.

13.2. Bourgain–Pavlović

Bourgain and Pavlović (2008) showed that the border case $\dot{B}_{\infty}^{-1, \infty}$ (a slightly larger space than BMO^{-1}) is ill-posed: there exist smooth initial data that are arbitrarily small in $\dot{B}_{\infty}^{-1, \infty}$ but produce solutions arbitrarily large in the same norm after arbitrarily short time—a phenomenon known as **norm inflation**.¹⁰

Thus BMO^{-1} is very close to the threshold for a good theory.

13.3. Germain–Pavlović–Staffilani

Germain, Pavlović, and Staffilani (2007) showed that the Koch–Tataru solution enjoys higher regularity: it is real analytic in space for every $t > 0$, with uniform decay estimates

$$\|\partial^\alpha u(\cdot, t)\|_{L^\infty} \leq C_\alpha t^{-\frac{|\alpha|+1}{2}}$$

for any multi-index α .¹¹

⁷J. T. Beale, T. Kato, and A. Majda, “Remarks on the breakdown of smooth solutions for the 3-D Euler equations,” *Comm. Math. Phys.* 94, 61–66 (1984). DOI Zbl

⁸P. Constantin and C. Fefferman, “Direction of vorticity and the problem of global regularity for the Navier-Stokes equations,” *Indiana Univ. Math. J.* 42, 775–789 (1993). DOI Zbl

⁹H. Koch and D. Tataru, “Well-posedness for the Navier-Stokes equations,” *Adv. Math.* 157, 22–35 (2001). DOI Zbl

¹⁰J. Bourgain and N. Pavlović, “Ill-posedness of the Navier-Stokes equations in a critical space in 3D,” *J. Funct. Anal.* 255, 2233–2247 (2008). DOI arXiv Zbl

¹¹P. Germain, N. Pavlović, and G. Staffilani, “Regularity of solutions to the Navier-Stokes equations evolving from small data in BMO^{-1} ,” *Int. Math. Res. Not.* 2007, rnm087 (2007). DOI arXiv Zbl

This implies, in particular, that solutions starting from small BMO^{-1} data become classical instantaneously.

So the picture is:

- at or below the critical threshold, one has good control;
- above it, the global theory becomes much more delicate.

14. Numerical blowup searches

Given the difficulty of the theoretical problem, it is natural to search numerically for singularity candidates.

14.1. Kerr's anti-parallel vortex tubes

In 1993, Kerr numerically studied anti-parallel vortex tubes and found evidence suggestive of finite-time blowup.

Later, Hou and Li revisited the same scenario with much higher resolution and found no blowup: the apparent singular behavior was a numerical artifact of insufficient resolution.

This illustrates a central challenge:

- quantities may grow rapidly and seem to diverge;
- but at later times they may deplete instead.

14.2. Other candidate scenarios

More complicated constructions, such as folded sheets and multiscale structures, have also been proposed as possible blowup mechanisms. But the equations repeatedly resist definitive numerical confirmation.

14.3. Core numerical difficulties

Numerically detecting blowup is intrinsically hard because:

1. if a true singularity is approached, resolution requirements become extreme;
2. truncation and discretization errors can dominate;
3. artificial numerical viscosity may suppress the effect one is trying to detect;
4. numerical evidence alone cannot establish a theorem.

15. Computer-assisted proofs

A more rigorous numerical paradigm is the **computer-assisted proof**.

15.1. General strategy

The idea is:

1. Compute a highly accurate approximate solution.
2. Show, by a fixed-point or perturbative argument, that a true solution exists near it.
3. Track all constants and errors rigorously using interval arithmetic.

Thus one upgrades a numerical candidate into a theorem.

15.2. Interval arithmetic

Instead of floating-point numbers, one propagates intervals guaranteed to contain the true values. This provides rigorous error bounds.

This methodology has become increasingly effective in PDE over the last decade, thanks to advances in software, hardware, and analysis.

16. Tao's perspective on the difficulty

Terence Tao emphasized that any proof of global regularity must go beyond standard energy methods.

16.1. Logarithmically supercritical models

Tao considered modified Navier–Stokes equations where the Laplacian $\nu\Delta$ is replaced by a dissipation operator D that is slightly stronger than $\nu\Delta$. Specifically, he proved global regularity for dissipation of the form $|\nabla|^{\frac{15}{8}}$ (which is more regularizing than Δ), and even for $|\nabla|^{\frac{5}{4}+\varepsilon}$ with a suitable logarithmic correction. The key threshold is that if the dissipation scales like $|\nabla|^s$ with $s > 1$ (this is the **hyperdissipative** regime), the problem becomes subcritical and global regularity can be proved for sufficiently large s .

This shows that just a bit more damping changes the problem dramatically.

16.2. Averaged Navier–Stokes and blowup models

Tao also constructed modified or averaged Navier–Stokes-type systems that preserve many soft features of the original equations—energy identity, Sobolev estimates, symmetries—yet admit finite-time blowup. In these models, the nonlinear term $(u \cdot \nabla)u$ is replaced by an averaged or truncated version that retains the same energy estimates and scaling properties but removes certain cancellations present in the genuine nonlinearity. The blowup is constructed via a self-similar ansatz combined with an Ornstein–Uhlenbeck-type stochastic construction.

Conclusion: soft properties alone cannot distinguish global regularity from blowup. One must use the precise structure of the genuine Navier–Stokes nonlinearity.

17. Onsager's conjecture and convex integration

The talk then turned to a different but related line of development: very rough solutions to Euler and Navier–Stokes.

17.1. Onsager's conjecture

For weak solutions of 3D Euler with Hölder continuity C^α (or, more generally, Besov regularity $B_{3,\infty}^\alpha$):

- if $\alpha > \frac{1}{3}$, energy is conserved;
- if $\alpha < \frac{1}{3}$, anomalous dissipation may occur, i.e., the time derivative $\frac{d}{dt} \int (\frac{1}{2}) |u|^2 dx$ may be nonzero.

The conservation part was proved by Constantin, E., and Titi: if $u \in L^3((0, T); B_{3,\infty}^\alpha)$ with $\alpha > \frac{1}{3}$, then the kinetic energy $\frac{1}{2} \|u(t)\|_{L^2}^2$ is constant in time.¹²

17.2. Convex integration

The flexibility part was developed through the method of **convex integration**, introduced into fluid dynamics by De Lellis and Székelyhidi, drawing on ideas of Nash and Gromov.

A sequence of works improved the regularity threshold, culminating in Isett's (2018) proof of Onsager's conjecture: for every $\alpha < \frac{1}{3}$, there exists a nonzero weak solution $u \in C_t C_x^\alpha$ of the 3D Euler equations that has compact support in time and therefore fails to conserve energy.¹³

The proof combines the method of convex integration with a **gluing approximation** technique using **Mikado flows** (introduced by Daneri and Székelyhidi). Later, Buckmaster, De Lellis,

¹²P. Constantin, W. E., and E. S. Titi, "Onsager's conjecture on the energy conservation for solutions of Euler's equation," **Comm. Math. Phys.** 165, 207–209 (1994). DOI

¹³P. Isett, "A proof of Onsager's conjecture," **Ann. of Math.** 188, 871–963 (2018). DOI arXiv Zbl

Székelyhidi, and Vicol strengthened this: for any prescribed nonnegative energy profile $e(t)$, there exists a weak solution of the 3D Euler equations whose kinetic energy equals $e(t)$ for almost every t .

Thus very rough Euler solutions can behave wildly.

18. Nonuniqueness for Navier–Stokes

A striking modern development is that rough weak solutions to 3D Navier–Stokes can be nonunique.

18.1. Buckmaster–Vicol

Buckmaster and Vicol (2019) proved nonuniqueness for weak solutions of 3D Navier–Stokes: there exist at least two distinct global weak solutions $u, v \in L_t^\infty L_x^2 L_t^2 \dot{H}^1$ with the same finite-energy initial data $u_0 \in L^2$. Moreover, these solutions can be chosen to be Hölder continuous with exponent $\alpha < \frac{1}{3}$ and to satisfy the energy equality (not just the inequality).¹⁴

The construction uses convex integration with **intermittent Beltrami flows**—a refinement of the Mikado flows used for Euler that incorporates a third scale parameter to control the nonlinear term at the level of the Navier–Stokes equations.

However, these solutions are too rough to be Leray–Hopf weak solutions (they do not satisfy the energy inequality). So this does not directly settle the classical uniqueness question for Leray–Hopf solutions.

18.2. Jia–Šverák program

Jia and Šverák (2014, 2015) proposed a program to prove nonuniqueness for Leray–Hopf solutions. They gave sufficient conditions for nonuniqueness in terms of the spectral properties of the linearized operator around a self-similar, scale-invariant solution of the Navier–Stokes equations. If the linearized operator has an unstable eigenvalue, one can construct a second solution lying on the unstable manifold, branching off from the self-similar background solution. The verification of these spectral conditions is in principle approachable by numerical simulation, since they involve only smooth functions.¹⁵

18.3. Albritton–Brue–Colombo

Albritton, Brué, and Colombo (2022) realized the Jia–Šverák program and proved nonuniqueness for forced Leray–Hopf solutions: there exist two distinct Leray–Hopf weak solutions of 3D Navier–Stokes, both with zero initial velocity $u_0 = 0$ and driven by the same smooth body force f , that coincide at $t = 0$ but differ at later times. The construction uses a self-similar, compactly supported vortex ring as the background solution and demonstrates that it is unstable under the Navier–Stokes dynamics in similarity variables.¹⁶

So with forcing, nonuniqueness at the Leray–Hopf level is known. The solutions live precisely on the borderline of the known well-posedness theory.

18.4. Announcement by Hou–Wang–Yang

In 2025, Hou, Wang, and Yang announced a computer-assisted proof of nonuniqueness for unforced Leray–Hopf solutions, building on the Jia–Šverák program. They construct a self-similar Leray–Hopf solution and rigorously certify the existence of an unstable eigenpair of the linearized operator

¹⁴T. Buckmaster and V. Vicol, “Nonuniqueness of weak solutions to the Navier–Stokes equation,” *Ann. of Math.* 189, 101–144 (2019). DOI arXiv Zbl

¹⁵H. Jia and V. Šverák, “Are the incompressible 3D Navier–Stokes equations locally ill-posed in the natural energy space?” *J. Funct. Anal.* 268, 3730–3766 (2015). DOI arXiv Zbl

¹⁶D. Albritton, E. Brué, and M. Colombo, “Non-uniqueness of Leray solutions of the forced Navier–Stokes equations,” *Ann. of Math.* 196, 415–455 (2022). DOI arXiv Zbl

by combining high-precision numerical computation with a decomposition of the operator into a coercive part plus a compact perturbation approximated by a finite-rank operator. The result establishes the existence of a second solution—indeed, infinitely many distinct Leray–Hopf solutions—for the same smooth, compactly supported initial data with zero external force.¹⁷

If fully confirmed, this would represent a major breakthrough: nonuniqueness at the Leray–Hopf level without external forcing.

19. Finite-time blowup for Euler with boundary

A landmark result concerns 3D incompressible Euler in a cylindrical domain.

19.1. Luo–Hou numerical scenario

In 2014, Luo and Hou numerically studied axisymmetric 3D Euler in a cylinder and observed strong vorticity growth near the boundary, suggesting a finite-time singularity.¹⁸

The geometry of the boundary appeared to play an essential role.

19.2. Chen–Hou theorem

In a series of papers beginning in 2023, Chen and Hou proved finite-time blowup for the 3D incompressible Euler equations in a cylindrical domain with no-slip boundary. Specifically, there exists smooth, finite-energy initial data (u_0, p_0) such that the corresponding solution of the 3D axisymmetric Euler equations develops a singularity in finite time. The velocity field remains $C^{1,\alpha}$ and has finite energy up to the singularity time.¹⁹

The proof is computer-assisted and follows the dynamic rescaling framework developed by Elgindi. The key steps are: (1) construct an approximate self-similar profile numerically; (2) linearize the rescaled equations around this profile and prove spectral stability; (3) upgrade to nonlinear stability via a fixed-point argument with rigorous error bounds using interval arithmetic.

19.3. Why this does not solve the Clay problem

This is a major theorem, but it does not resolve the Clay problem because:

- it concerns Euler, not Navier–Stokes;
- it takes place in a bounded domain with boundary;
- the boundary is crucial to the blowup mechanism.

So it does not answer the whole-space or periodic 3D Navier–Stokes regularity question.

20. Other blowup scenarios and dimension as a parameter

The lecture also mentioned numerical studies by Hou suggesting possible blowup scenarios for Navier–Stokes, including tornado-type structures with large vorticity amplification.

Another interesting idea is to treat the spatial dimension as a parameter in a generalized model. Numerically, one may find self-similar singularities in a noninteger dimension, such as (3.188). This suggests that 3D may lie near a threshold, and that one missing cancellation might separate regularity from blowup.

This is suggestive, not definitive.

¹⁷T. Hou, Y. Wang, and C. Yang, “Nonuniqueness of Leray–Hopf solutions to the unforced incompressible 3D Navier–Stokes equation,” arXiv:2509.25116 (2025). arXiv

¹⁸G. Luo and T. Y. Hou, “Potentially singular solutions of the 3D axisymmetric Euler equations,” *PNAS* 111, 12968–12973 (2014). DOI arXiv Zbl

¹⁹J. Chen and T. Y. Hou, “Finite time blowup of 2D Boussinesq and 3D Euler equations with $C^{1,\alpha}$ velocity and boundary,” *Ann. PDE* 9, 14 (2023). arXiv Zbl see also J. Chen and T. Y. Hou, *PNAS* 122, e2500940122 (2025). DOI

21. AI and neural networks in singularity discovery

The lecture then discussed neural-network-based discovery methods.

21.1. Neural networks as nonlinear ansätze

A neural network represents a function by composing affine maps with nonlinear activation functions, for example ReLU. Because this is a highly nonlinear parametrization, such networks can approximate complicated structures with relatively few parameters.

21.2. Physics-informed neural networks

In a physics-informed neural network (PINN), one minimizes a loss function based on the PDE residual and possibly its derivatives or constraints.

Thus the PDE itself guides the search for candidate solutions.

21.3. Applications to singularity discovery

The speaker described joint work using neural-network-based methods to discover self-similar blowup profiles for equations related to fluid dynamics, such as:

- axisymmetric 3D Euler models,
- the CCF equation,
- incompressible porous media equations.

These methods are good at discovering candidate singularity profiles, but they do not by themselves prove anything about Navier–Stokes.

22. Self-similar singularities

A major conceptual motif is the search for self-similar blowup.

22.1. Self-similar ansatz

Suppose blowup occurs at time T . One seeks solutions of the form

$$u(x, t) = \frac{1}{(T - t)^\beta} U\left(\frac{x - x_0}{(T - t)^\gamma}\right)$$

or in a more general time-dependent rescaled form. For the Navier–Stokes equations, scaling invariance forces the relation

$$2\beta + 1 = \gamma.$$

Here:

- β is the amplitude blowup exponent,
- γ is the spatial concentration exponent,
- U is the profile. The energy-critical case corresponds to $\beta = \frac{1}{2}$, $\gamma = 1$ (the Leray self-similar scaling), for which U is expected to belong to L^3 .

Under suitable rescaling variables

$$\xi = \frac{x - x_0}{(T - t)^\gamma} \quad \tau = -\log(T - t)$$

a self-similar singularity corresponds to convergence to a stationary profile in (ξ, τ) -coordinates.

This is attractive because the complicated singular behavior in physical variables becomes a fixed-point problem in rescaled variables.

22.2. Program for proving blowup

A possible route to a theorem is:

1. Find an approximate self-similar profile numerically.
2. Linearize around it.
3. Prove good spectral or stability properties.
4. Upgrade this to nonlinear control.
5. Use a computer-assisted argument to make every estimate rigorous.
6. Translate back to physical variables and conclude blowup.

22.3. Restrictions on self-similarity

Not every self-similar ansatz is admissible.

Caffarelli, in earlier lectures on the Millennium problems, emphasized that one cannot expect the simplest self-similar Navier–Stokes singularities directly, due to exclusion results such as those of Nečas, Růžička, and Šverák (1996). They proved that there are no nontrivial self-similar solutions of the form

$$u(x, t) = \frac{1}{(T - t)^{\frac{1}{2}}} U\left(\frac{x}{(T - t)^{\frac{1}{2}}}\right)$$

with $U \in L^3(\mathbb{R}^3)$ (the natural energy space for the profile). More precisely, if U is divergence-free and $U \in L^3(\mathbb{R}^3)$, then the only solution of this self-similar form is $U \equiv 0$.²⁰

This rules out the most naive self-similar blowup scenario for Navier–Stokes.

Hence, if one seeks singularity for Navier–Stokes, one may need first to understand Euler-type singularity formation and then show viscosity is lower order near the singularity.

Recent work of Constantin, Ignatova, and Vicol also places restrictions on possible self-similar exponents for Euler. The lecture stated, roughly, that one must have

$$\gamma \geq \frac{2}{5}$$

in general, and in the axisymmetric case

$$\gamma \geq \frac{1}{2}.$$

These constraints matter because they interact with the Navier–Stokes scaling.

23. Blowup from rough data for Euler

Another direction is to lower the regularity of the initial data.

23.1. Elgindi’s theorem

Elgindi (2021) proved finite-time blowup for 3D Euler in the whole space $\mathbb{R}^3$ (without boundaries). Starting from initial data $u_0 \in C^{1,\alpha}(\mathbb{R}^3)$ (Hölder continuous gradient) that is axisymmetric and has no swirl, the corresponding solution develops a singularity at a finite time $T < \infty$. The vorticity blows up like $\omega(t) \approx (T - t)^{-1}$ as $t \rightarrow T^-$, and the velocity gradient blows up like $\nabla u \approx (T - t)^{-1}$.²¹

²⁰J. Nečas, M. Růžička, and V. Šverák, “On Leray’s self-similar solutions of the Navier-Stokes equations,” **Acta Math.** 176, 283–294 (1996). DOI Zbl

²¹T. M. Elgindi, “Finite-time singularity formation for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$,” **Ann. of Math.** 194, 647–727 (2021). DOI arXiv Zbl

The proof uses a dynamic rescaling formulation: the solution is written in self-similar variables, and the problem is reduced to proving convergence to a stable, nontrivial stationary profile of the rescaled equations.

This was a major breakthrough: it showed singularity formation in the whole space without boundaries, though not from smooth data (the initial velocity is $C^{1,\alpha}$ but not C^2).

23.2. Stability and related works

This was further developed by Elgindi, Ghoul, and Masmoudi, who studied stability of the profile.

Other related works used multiscale constructions and analogous mechanisms.

An important open question remains:

Can one upgrade these rough-data blowup constructions to smooth-data blowup?

24. Where we stand

The landscape today is far richer than it was 25 years ago.

We now know:

- global existence of Leray–Hopf weak solutions;
- complete global theory in 2D;
- strong partial regularity theory;
- many conditional regularity criteria;
- nonuniqueness for very rough Navier–Stokes solutions;
- nonuniqueness for forced Leray–Hopf solutions;
- major announcements toward unforced Leray–Hopf nonuniqueness;
- finite-time blowup for 3D Euler in a cylinder;
- finite-time blowup for 3D Euler from rough initial data in the whole space;
- new numerical and AI-assisted tools for discovering candidate singularities.

But we still do not know:

- whether smooth 3D Navier–Stokes solutions can blow up in finite time;
- whether smooth solutions always exist globally;
- whether Leray–Hopf weak solutions are unique in the classical unforced setting.

25. Outlook

According to the lecture, several broad future directions appear plausible:

1. Global regularity via new structure

Any proof must go beyond energy methods and exploit deep structure specific to Navier–Stokes.

2. Singularity formation through related models

Euler and modified equations may reveal the mechanisms needed for Navier–Stokes.

3. Leray–Hopf nonuniqueness and weak solution theory

New notions of solution, or refined understanding of old ones, may reshape the problem.

4. Computer-assisted proofs

These are becoming increasingly central.

5. AI-assisted discovery

While still exploratory, such methods may help identify candidate structures that can later be proved rigorously.

The central question remains:

Do there exist smooth initial data for 3D incompressible Navier–Stokes that produce a finite-time singularity, or are smooth solutions always global?

After more than 200 years of study, this remains unknown.

26. Clean mathematical summary

For reference, here is a concise statement of the main mathematical objects mentioned.

26.1. 3D incompressible Navier–Stokes

On $\mathbb{R}^3$ or $\mathbb{T}^3$,

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u + f$$

$$\nabla \cdot u = 0$$

$$u(\cdot, 0) = u_0.$$

26.2. 3D incompressible Euler

Set $\nu = 0$:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = f \quad \nabla \cdot u = 0.$$

26.3. Vorticity

$$\omega = \nabla \times u.$$

In 3D Navier–Stokes,

$$\partial_t \omega + u \cdot \nabla \omega = \omega \cdot \nabla u + \nu \Delta \omega.$$

In 2D Navier–Stokes,

$$\partial_t \omega + u \cdot \nabla \omega = \nu \Delta \omega.$$

26.4. Energy inequality for Leray–Hopf solutions

$$\frac{1}{2} \|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds \leq \frac{1}{2} \|u_0\|_{L^2}^2.$$

26.5. Prodi–Serrin criterion

If

$$u \in L_t^p L_x^q \quad \frac{2}{p} + \frac{3}{q} \leq 1 \quad q > 3$$

then u is smooth.

26.6. Beale–Kato–Majda criterion

If a smooth solution blows up at time T , then

$$\int_0^T \|\omega(\cdot, t)\|_{L^\infty} dt = \infty.$$